THE QUADRATIC TANG–ZHANG REFINEMENT HOLDS FOR EVERY DEGREE $n \geq 10^{200000}$

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Abstract. Tang and Zhang conjectured that if a polynomial of degree n has all its zeros in the closed unit disk, a is one of its zeros, and $w_1, \ldots, w_{n-1}$ are its critical points, then

$$\sum_{j=1}^{n-1} |a - w_j|^{-\lambda} \geq n - 1 \qquad (\lambda \geq 1).$$

We make the high-degree argument for the quadratic case completely effective. More precisely, we prove the conjectured inequality for $\lambda = 2$ whenever $n \geq 10^{200000}$; consequently it also holds for every $\lambda \geq 2$ in this range. The proof starts from the polar and origin identities isolated in Tao's digestion of the proof of Sendov's conjecture. It divides the parameter $\delta = 1 - a^2$ into the explicit regimes $\delta \leq 1/\log(n-1)$ and $\delta \geq 1/\log(n-1)$. Near the boundary, a division-free centered expansion of the first origin integral is supplied with an absolute remainder bound, while the polar inequality yields the decisive factor-three variance estimate. In the complementary regime, an explicit polar endpoint gap controls both terms in the exact derivative identity. All asymptotic notation is eliminated, and a numerical ledger records the elementary inequalities used at the stated threshold.

Generative-AI disclosure. This manuscript was prepared by guiding ChatGPT 5.6 Sol Pro Chat to apply the effectivization strategy of Zhang's paper [Zha26] to the two-regime proof contained in the AI-generated working notes [AI26], with the aim of replacing an unspecified "sufficiently large degree" by an explicit admissible value of N. The source supplied to ChatGPT for that two-regime argument was the PDF *The quadratic Tang–Zhang refinement in high degree: A two-regime working proof* [AI26]. The article, including its mathematical exposition and bibliography, was completed with AI assistance. The author made many modifications to the generated manuscript and assumes responsibility for the final version.

1. Introduction

Let p be a complex polynomial of degree $n \geq 2$, all of whose zeros lie in the closed unit disk $\overline{\mathbb{D}}$. Sendov's conjecture asserts that, for every zero a of p, at least one critical point w satisfies $|a - w| \leq 1$. The conjecture was verified on the boundary by Rubinstein [Rub68], in degrees at most five by Meir and Sharma [MS69, pp. 459–467], and ultimately in degrees at most eight by Brown and Xiang [BX99]. Earlier asymptotic, high-degree, and near-boundary results include [BRS85, Deg14, Chi11, Cha20]. Tao proved that it holds in every sufficiently high degree [Tao22, Theorem 1.2]. An explicit implementation of Tao's high-degree theorem, with the universal threshold 10^{200000}, was subsequently given in [Zha26, Theorem 1.2 and pp. 27–28]. A computer-assisted proof candidate for the full conjecture, accompanied by exact certificates and a separate Lean formalization, has been released for independent review in [Maz26, pp. 1–19]; Tao's later digestion isolates four communication identities and a two-parameter feasibility mechanism [Tao26, Propositions 10–11].

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Tang and Zhang proposed the following strengthening [TZ25, p. 5, Conjecture 1.10]. In the notation below, the convention is that a summand is $+\infty$ when $w_j = a$.

Conjecture 1.1. *Let p be a polynomial of degree $n \geq 2$, all of whose zeros lie in $\overline{\mathbb{D}}$. If a is a zero of p and $w_1, \ldots, w_{n-1}$ are the zeros of p', counted with multiplicity, then*

$$\sum_{j=1}^{n-1} \frac{1}{|a - w_j|^\lambda} \geq n - 1 \qquad (\lambda \geq 1).$$

As $\lambda \to \infty$, this recovers Sendov's conjecture. The endpoint $\lambda = 1$ is the strongest member of the family in the sense of power means, while the case $\lambda = 2$ is the first one that admits a genuinely quadratic averaged obstruction. The purpose of this paper is to replace the phrase "sufficiently large degree" in the two-regime quadratic argument by a completely explicit integer.

Set

$$N_{\text{quad}} := 10^{200000}. \tag{1.1}$$

The main result is the following.

Theorem 1.2. *Let p be a complex polynomial of degree $n \geq N_{\text{quad}}$, all of whose zeros lie in $\overline{\mathbb{D}}$. If a is a zero of p and $w_1, \ldots, w_{n-1}$ are the critical points of p, counted with multiplicity, then*

$$\sum_{j=1}^{n-1} \frac{1}{|a - w_j|^2} \geq n - 1. \tag{1.2}$$

The two-regime architecture used below is adapted from the working notes [AI26]; the present manuscript supplies the explicit constant audit and the threshold in (1.1).

The power-mean inequality immediately extends the conclusion to all larger exponents.

Corollary 1.3. *Under the assumptions of Theorem 1.2, one has*

$$\sum_{j=1}^{n-1} \frac{1}{|a - w_j|^\lambda} \geq n - 1 \qquad (\lambda \geq 2).$$

The proof uses only elementary polynomial identities, the arithmetic–geometric mean inequality, Cauchy's coefficient estimate, and one scalar exponential comparison. The threshold in (1.1) is deliberately not optimized. Its role is to make every remainder estimate transparent and to parallel the explicit-constant philosophy of [Zha26, pp. 3–4 and Appendix A].

We now outline the argument. After rotation and translation, the distinguished zero is placed at the origin and the original disk becomes $\{z : |z + a| \leq 1\}$ with $0 \leq a \leq 1$. A hypothetical counterexample gives reciprocal critical-point coordinates q_j satisfying the averaged inequality

$$s := \frac{1}{m} \sum_{j=1}^{m} |q_j|^2 < 1, \qquad m = n - 1.$$

The polar identity forces a relationship between

$$\alpha := \frac{m}{2}(1 - a^2) \quad \text{and} \quad \beta := 1 - 2ax + a^2, \qquad x := \operatorname{Re} \frac{1}{m} \sum_j q_j,$$

namely

$$0 < \beta < \frac{\alpha}{3 + \alpha}.$$

Near the unit circle, where $1-a^2 \le 1/\log m$, this gives a factor-three control of the centered quadratic variance. A quantitative centered product expansion then shows that the first origin integral has modulus strictly larger than one. Away from the unit circle, the polar integral creates an explicit endpoint gap of size at least $\log m/m$, and the exact derivative identity makes the origin integral asymptotic with an error smaller than $1-a$. The two regimes meet at the explicit seam $1-a^2 = 1/\log m$.

2. Normalization and communication identities

We first pass to the coordinate system used throughout the proof. After rotating the original polynomial, we may suppose that its distinguished zero is the real number $a \in [0,1]$. Translating by a, and suppressing an irrelevant nonzero scalar factor, gives

$$P(z) = z\prod_{i=1}^{m}(z-z_i), \qquad |z_i + a| \le 1, \qquad m = n-1.$$

The critical points of P are the translated critical points of the original polynomial. If one of them is zero, then the desired inequality is immediate. We therefore suppose that all critical points are nonzero and write them as

$$w_j = -\frac{1}{q_j}, \qquad 1 \le j \le m.$$

A counterexample to (1.2) is exactly the strict averaged obstruction

$$s := \frac{1}{m}\sum_{j=1}^{m}|q_j|^2 < 1. \tag{2.1}$$

We also put

$$\mu := \frac{1}{m}\sum_{j=1}^{m} q_j = x+iy, \qquad \delta := 1-a^2, \qquad \alpha := \frac{m\delta}{2},$$

$$\beta_s(t) := 1 - 2atx + a^2st^2, \qquad \beta(t) := 1-2atx+a^2t^2, \qquad \beta := \beta(1).$$

Finally, define

$$F(t) := \prod_{j=1}^{m}(1-atq_j), \qquad Z := \prod_{i=1}^{m}(a+z_i), \qquad Q := \prod_{j=1}^{m} q_j, \qquad J := ZQ.$$

The disk condition and (2.1) give

$$|Z| \le 1, \qquad |Q|^2 \le s^m < 1, \qquad |J| < 1. \tag{2.2}$$

The rest of the proof derives the contradictory inequality $|J| > 1$.

The first lemma records the two origin identities needed below. They are the shifted forms of the identities isolated in [Tao26, equations (9)–(10)].

Lemma 2.1. *Assume $a > 0$. Then*

$$(-1)^m J = (m+1)\int_0^1 F(t)\,\mathrm{d}t. \tag{2.3}$$

Moreover,

$$F'(t) = -ma\mu F(t) - a^2 t\sum_{j=1}^{m} q_j^2 \prod_{k\neq j}(1-atq_k). \tag{2.4}$$

If

$$R := a^2 \int_0^1 t \sum_{j=1}^m q_j^2 \prod_{k \neq j} (1 - atq_k)\, \mathrm{d}t,$$

then

$$1 = F(1) + ma\mu \int_0^1 F(t)\, \mathrm{d}t + R \tag{2.5}$$

and

$$|R| \leq \mathrm{e}^{1/2} a^2 m \int_0^1 t\, \beta_s(t)^{(m-1)/2}\, \mathrm{d}t. \tag{2.6}$$

Proof. Since the zeros of P' are $-1/q_j$ and the leading coefficient of P' is $m+1$,

$$P'(z) = \frac{m+1}{Q} \prod_{j=1}^m (1 + zq_j).$$

Integrating from $-a$ to 0, and using

$$P(-a) = (-1)^{m+1} aZ,$$

gives (2.3). Differentiating the product defining F and using $\sum_j q_j = m\mu$ gives (2.4); integration yields (2.5).

For the remainder estimate, put $b_k(t) = |1 - atq_k|$. The arithmetic–geometric mean inequality gives, for every j,

$$\prod_{k \neq j} b_k(t) \leq \left(\frac{\sum_{k \neq j} b_k(t)^2}{m-1} \right)^{(m-1)/2} \leq \left(\frac{m\beta_s(t)}{m-1} \right)^{(m-1)/2}.$$

Since $\sum_j |q_j|^2 = ms < m$ and

$$\left(\frac{m}{m-1} \right)^{(m-1)/2} \leq \mathrm{e}^{1/2},$$

we obtain (2.6). □

The polar identity is the second communication mechanism. Its division-free form is implicit in [Maz26, Section 5] and is written explicitly in [Tao26, Proposition 10].

Lemma 2.2. *If $0 < a < 1$, then*

$$\prod_{i=1}^m \frac{1 - a(a + z_i)}{a - (a + z_i)} = \int_0^1 \prod_{j=1}^m \big(a + t(1 - a^2) q_j\big)\, \mathrm{d}t, \tag{2.7}$$

where removable singularities are understood by cancellation. Consequently,

$$1 \leq \int_0^1 \left(a^2 + 2a\delta x t + \delta^2 s t^2\right)^{m/2} \mathrm{d}t. \tag{2.8}$$

Proof. The identity follows by integrating the factorization of P' from 0 to δ/a and comparing the result with the zero factorization of $P(\delta/a)$; this is the shifted calculation in [Tao26, equation (8)]. For $u = a + z_i$ with $|u| \leq 1$,

$$|1 - au|^2 - |a - u|^2 = (1 - a^2)(1 - |u|^2) \geq 0.$$

Thus the modulus of the left-hand side of (2.7) is at least one. The triangle inequality, followed by the quadratic-mean inequality, gives

$$\left|\prod_{j=1}^{m}(a+t\delta q_j)\right| \leq \left(\frac{1}{m}\sum_{j=1}^{m}|a+t\delta q_j|^2\right)^{m/2},$$

and the average inside the last display is exactly the quadratic in (2.8). □

The following scalar inequality is the exponential comparison used to simplify the polar estimate. A fully algebraic proof is given in [Maz26, pp. 5–6, Lemma 5.1].

Lemma 2.3. *For every $u > 0$,*

$$\exp\left(\frac{3u}{u+3}\right) - 1 - \frac{3u}{u+3} < u - 1 + \mathrm{e}^{-u}.$$

We may now extract the precise polar information used in both regimes.

Proposition 2.4. *Every counterexample with $0 < a < 1$ satisfies*

$$0 < \beta < \frac{\alpha}{3+\alpha} < 1. \tag{2.9}$$

In particular,

$$x > \frac{a}{2} > 0. \tag{2.10}$$

Proof. Since $s < 1$, $t^2 \leq t$, and

$$2ax + \delta = 2 - \beta,$$

the integrand in (2.8) is strictly smaller, except at an endpoint of measure zero, than

$$\exp\bigl(\alpha[-1 + (2-\beta)t]\bigr).$$

Hence

$$1 < \int_0^1 \exp\bigl(\alpha[-1+(2-\beta)t]\bigr)\,\mathrm{d}t. \tag{2.11}$$

This inequality first forces $\beta < 1$: if $\beta \geq 1$, then the exponent is nonpositive on $[0,1]$ and is negative on a set of positive measure, so the integral is smaller than one. Thus $\alpha(1-\beta) \geq 0$. If $\beta \geq \alpha/(3+\alpha)$ and $u := \alpha(1-\beta)$, then

$$0 \leq u \leq \frac{3\alpha}{3+\alpha}.$$

The function $v \mapsto \mathrm{e}^v - 1 - v$ is increasing for $v \geq 0$, so Lemma 2.3 gives

$$\mathrm{e}^u - 1 - u < \alpha - 1 + \mathrm{e}^{-\alpha}.$$

Equivalently,

$$\mathrm{e}^u - \mathrm{e}^{-\alpha} < \alpha + u = \alpha(2-\beta),$$

which contradicts the exact evaluation of (2.11). Thus the upper bound in (2.9) holds. Since $s < 1$, one has $|x| \leq |\mu| < 1$. The identity

$$\beta = 1 - x^2 + (x-a)^2$$

therefore shows that $\beta > 0$. Finally,

$$1 - \beta = 2a\left(x - \frac{a}{2}\right) > 0,$$

which proves (2.10). □

3. A quantitative centered product expansion

The near-boundary argument requires an expansion that retains the quadratic centered term and controls all remaining terms uniformly under the averaged hypothesis (2.1). Put

$$r_j := q_j - \mu, \qquad v := \frac{1}{m}\sum_{j=1}^{m} |r_j|^2 = s - |\mu|^2, \qquad \kappa := \frac{1}{m}\sum_{j=1}^{m} r_j^2.$$

Then

$$\sum_{j=1}^{m} r_j = 0, \qquad |\kappa| \le v.$$

The polar estimate supplies the more important inequalities

$$v \le \beta, \qquad |\mu|^2 \ge x^2 \ge 1-\beta, \qquad \frac{v}{|\mu|^2} < \frac{\alpha}{3}. \tag{3.1}$$

Indeed, $v \le 1 - |\mu|^2 \le 1 - x^2 \le \beta$, and the last inequality follows from (2.9).

We next give a fully quantitative version of the centered expansion. The constants are intentionally rounded upward.

Lemma 3.1. *Assume $m \ge 100$, $a \ge 9/10$, $x \ge 2/5$, $x > a/2$, and $s \le 1$. With the notation above, let*

$$I := \int_0^1 F(t)\,\mathrm{d}t, \qquad p := \frac{m-2}{2}.$$

Then

$$\left|(m+1)I - \frac{1}{a\mu}\left(1 - \frac{\kappa}{(m-1)\mu^2}\right)\right| \le \frac{\mathcal{E}}{a|\mu|}, \tag{3.2}$$

where

$$\mathcal{E} := 2\cdot 10^4 \frac{v^{3/2}}{m^{3/2}} + 20m^2 \mathrm{e}^{-m/200}\mathbb{1}_{\{v>1/16\}} + 100m^2\beta^p. \tag{3.3}$$

Proof. For $u \in \mathbb{C}$, define

$$G_u(t) := \prod_{j=1}^{m} \big(1 - at(\mu + ur_j)\big).$$

As a polynomial in u,

$$\begin{aligned} G_0(t) &= (1-at\mu)^m, \qquad \partial_u G_u(t)|_{u=0} = 0, \\ \frac{1}{2}\partial_u^2 G_u(t)|_{u=0} &= -\frac{ma^2\kappa t^2}{2}(1-at\mu)^{m-2}. \end{aligned} \tag{3.4}$$

Write the difference between G_1 and the three displayed Taylor terms as $\mathcal{R}_3(t)$.

We first estimate $\mathcal{R}_3$ on the decreasing part of the square-mean majorant. Let

$$t_0 := \frac{x}{as}, \qquad t_c := \frac{x}{8av},$$

with the convention $t_c = +\infty$ when $v = 0$. For

$$0 < t \le \min(1, t_0, t_c), \qquad R_t := \left(\frac{x}{2atv}\right)^{1/2} \ge 2.$$

The elementary inequality $|1+z| \le \exp(\operatorname{Re} z + |z|^2/2)$ and the cancellation $\sum_j r_j = 0$ yield, on $|u| = R_t$,

$$\begin{aligned} |G_u(t)| &\le \exp\left(-matx + \frac{ma^2t^2}{2}\big(|\mu|^2 + |u|^2 v\big)\right) \\ &\le \exp(-matx/4). \end{aligned}$$

Here $t \le t_0$ gives $ats \le x$. Cauchy's coefficient estimate, summed from degree three onward, therefore gives

$$|\mathcal{R}_3(t)| \le 2R_t^{-3}\mathrm{e}^{-matx/4} = 2^{5/2}\left(\frac{atv}{x}\right)^{3/2}\mathrm{e}^{-matx/4}.$$

Integrating to infinity and using $\Gamma(5/2) = 3\sqrt{\pi}/4$ gives

$$a|\mu|(m+1)\int_0^{\min(1,t_0,t_c)} |\mathcal{R}_3(t)|\,\mathrm{d}t \le 2\cdot 10^4\frac{v^{3/2}}{m^{3/2}}.$$

The numerical constant follows from $a \ge 9/10$, $x \ge 2/5$, $|\mu| \le 1$, and $m+1 \le 2m$.

Suppose next that $t_c < \min(1,t_0)$. Then $v > x/(8a) > 1/16$. On the interval $[t_c, \min(1,t_0)]$ one has

$$\beta_s(t) \le 1 - axt \le \mathrm{e}^{-axt}, \qquad \frac{maxt}{2} \ge \frac{mx^2}{16v} \ge \frac{m}{100}.$$

The same majorant applies to $|G_0(t)|$. For the quadratic term in (3.4), the exponent is $(m-2)axt/2$, which is at least $m/200$ for $m \ge 100$, and the remaining coefficient is at most $m/2$. Consequently, after multiplying by $a|\mu|(m+1)$, the total contribution of G_1, G_0, and the quadratic term on this interval is at most

$$20m^2\mathrm{e}^{-m/200}\mathbb{1}_{\{v>1/16\}}.$$

If $t_0 < 1$, then β_s is increasing on $[t_0, 1]$ and

$$\beta_s(t) \le \beta_s(1) \le \beta.$$

The quadratic-mean inequality gives $|G_1(t)| \le \beta^{m/2}$, and the same bound with the appropriate exponent holds for the two Taylor terms. Their total contribution, after the same normalization, is at most

$$40m^2\beta^p. \tag{3.5}$$

It remains to compare the integrals of the first two Taylor terms with their limiting beta integrals. Direct integration gives

$$(m+1)\int_0^1 (1-at\mu)^m\,\mathrm{d}t = \frac{1-(1-a\mu)^{m+1}}{a\mu}.$$

Since $|1-a\mu|^2 \le \beta_s(1) \le \beta$, the omitted endpoint has normalized modulus at most $\beta^{(m+1)/2}$.

For the quadratic term, the substitution $u = 1 - a\mu t$ yields

$$\int_0^1 t^2(1-a\mu t)^{m-2}\,\mathrm{d}t = \frac{1}{(a\mu)^3}\Bigg[\frac{2}{m(m-1)(m+1)} - (1-a\mu)^{m-1}\left(\frac{1}{m-1} - \frac{2(1-a\mu)}{m} + \frac{(1-a\mu)^2}{m+1}\right)\Bigg].$$

Thus its principal contribution is

$$-\frac{1}{a\mu}\frac{\kappa}{(m-1)\mu^2},$$

and its normalized endpoint error is at most $30m\beta^{(m-1)/2}$. Combining this estimate with (3.5), and enlarging the coefficient to $100m^2$, proves (3.2)–(3.3). $\square$

The next lemma is the numerical audit that converts the explicit remainder into the scale required at the seam $\delta = 1/\log m$.

Lemma 3.2. *Let*
$$M_\star := 10^{199999}, \qquad L := \log m.$$
Suppose $m \ge M_\star$ and
$$0 < \alpha \le \frac{m}{2L}, \qquad 0 \le v \le \min\left(1, \frac{\alpha}{3}\right), \qquad 0 < \beta \le \frac{\alpha}{3+\alpha}.$$
Then the quantity in (3.3) satisfies
$$\mathcal{E} \le \frac{\alpha}{50m}. \tag{3.6}$$

Proof. At the threshold $m \ge M_\star$ one has
$$\begin{gathered} L > 4.6 \cdot 10^5, \qquad 3 < \frac{m}{100L}, \\ \frac{2 \cdot 10^4}{\sqrt{m}} < \frac{1}{150}, \qquad \frac{100}{\sqrt{m}} < \frac{1}{600L}, \\ 15000 m^3 3^{-(m-2)/2} < 1, \qquad 15000 < m. \end{gathered} \tag{3.7}$$
These inequalities follow immediately after taking logarithms.

For the first term in (3.3), use $v^{3/2} \le \alpha$ when $\alpha \le 1$, and use $v \le 1 \le \alpha$ when $\alpha \ge 1$. The third inequality in (3.7) then gives a contribution at most $\alpha/(150m)$.

If the middle indicator is nonzero, then $v > 1/16$. Since $v \le \beta < \alpha/(3+\alpha)$, it follows that $\alpha > 1/5$. The elementary inequality
$$15000 m^3 \mathrm{e}^{-m/200} < 1$$
therefore bounds the middle term by $\alpha/(150m)$.

It remains to estimate the last term. Put $p = (m-2)/2$. If $0 < \alpha \le 1$, then
$$\beta^p \le \left(\frac{\alpha}{3}\right)^p \le \alpha 3^{-p},$$
and the corresponding inequality in (3.7) gives the required bound. If $1 \le \alpha \le m/(4L)$, then
$$3 + \alpha \le \frac{m}{3L}, \qquad \beta^p \le \exp\left(-\frac{3p}{3+\alpha}\right) \le m^{-4}.$$
Hence $100m^2\beta^p \le 100m^{-2} \le 1/(150m) \le \alpha/(150m)$, by (3.7). If $m/(4L) \le \alpha \le m/(2L)$, then
$$3 + \alpha \le \frac{51m}{100L}, \qquad \beta^p \le m^{-5/2}.$$
In the final range, $\alpha/(150m) \ge 1/(600L)$, so the fourth inequality in (3.7) applies. Thus the last term is also at most $\alpha/(150m)$. Adding the three estimates gives (3.6). □

4. The near-boundary regime

We now exclude the first explicit regime.

Proposition 4.1. *There is no counterexample satisfying*
$$m \ge M_\star, \qquad 0 < \delta \le \frac{1}{\log m}.$$

Proof. The assumptions give
$$\alpha = \frac{m\delta}{2} \le \frac{m}{2\log m}.$$

Moreover, $a^2 = 1 - \delta > 9/10$, and Proposition 2.4 gives $x > a/2 > 2/5$. Thus Lemma 3.1 applies. Equations (3.1) and Lemma 3.2 yield

$$(-1)^m J = \frac{1}{a\mu}\left(1 - \frac{\kappa}{(m-1)\mu^2} + E\right), \qquad |E| \le \frac{\alpha}{50m}. \tag{4.1}$$

Put

$$r := \frac{|\kappa|}{(m-1)|\mu|^2} + |E|.$$

By (3.1),

$$r < \frac{\alpha}{3(m-1)} + \frac{\alpha}{50m} < \frac{2\alpha}{5m} \le \frac{1}{5\log m}. \tag{4.2}$$

In particular, $r < 1/2$. Taking logarithms of absolute values in (4.1), and using $\log(1-r) \ge -r/(1-r)$, gives

$$\begin{aligned}\log|J| &\ge -\log a - \log|\mu| + \log(1-r)\\ &\ge -\log a - \frac{r}{1-r}.\end{aligned}$$

Since $a^2 = 1 - 2\alpha/m$,

$$-\log a = -\frac{1}{2}\log\left(1 - \frac{2\alpha}{m}\right) \ge \frac{\alpha}{m}.$$

The bound (4.2), together with $\log m > 4.6 \cdot 10^5$, gives $r/(1-r) < \alpha/(2m)$. Hence

$$\log|J| > \frac{\alpha}{2m} > 0.$$

This contradicts (2.2). □

5. The complementary regime

We next treat the range in which the distinguished zero is separated from the unit circle at the explicit logarithmic scale. The starting point is a chord estimate for the raw polar integral.

Lemma 5.1. *Let*

$$\rho := s - \beta_s(1) = \delta s - 1 + 2ax, \qquad N := \frac{m}{2} + 1.$$

Then

$$(1 + \delta\rho)^N \ge N\delta. \tag{5.1}$$

Whenever $N\delta > 1$, one has

$$\rho \ge \frac{\log(N\delta)}{N\delta} > 0. \tag{5.2}$$

Moreover,

$$\beta_s(1) \le 1 - \rho, \qquad \rho \le 2ax, \qquad x^2 > \frac{\rho}{4}.$$

Proof. The quadratic in (2.8) is

$$H(t) = a^2 + 2a\delta x t + \delta^2 s t^2.$$

It is convex, and its endpoint values are

$$H(0) = a^2, \qquad H(1) = 1 + \delta\rho.$$

Hence H lies below the chord

$$\ell(t) = a^2 + \delta(1+\rho)t.$$

If $\rho \le 0$, then both endpoint values of the chord are at most one, and the polar integral is strictly smaller than one, a contradiction. Thus $\rho > 0$. Direct integration gives

$$1 \le \int_0^1 \ell(t)^{m/2}\,\mathrm{d}t = \frac{(1+\delta\rho)^N - a^{2N}}{N\delta(1+\rho)},$$

which implies (5.1). Taking logarithms and using $\log(1+u) \le u$ proves (5.2).

Since $s < 1$, one has $\beta_s(1) = s - \rho \le 1 - \rho$. Also

$$\rho = \delta s - 1 + 2ax \le \delta - 1 + 2ax \le 2ax.$$

Finally, $x > a/2$ gives $x^2 > ax/2 \ge \rho/4$. $\square$

The endpoint gap makes the exact derivative identity effective. The following estimate contains all constants used in the complementary regime.

Lemma 5.2. *Suppose*

$$m \ge M_\star, \qquad \delta \ge \frac{1}{\log m}, \qquad 0 < a < 1.$$

Then

$$|F(1)| + |R| < \frac{\delta}{100}. \tag{5.3}$$

Proof. Put $L = \log m$ and $A = N\delta$. The assumptions imply

$$A \ge \frac{m}{2L}, \qquad \log A > \frac{L}{2}.$$

By Lemma 5.1,

$$|F(1)| \le \beta_s(1)^{m/2} \le \mathrm{e}^{-m\rho/2} \le A^{-1/(2\delta)} \le A^{-1/2} \le \sqrt{\frac{2L}{m}}.$$

At $m \ge M_\star$, the last expression is smaller than $\delta/1000$.

We next use (2.6). Put $r = (m-1)/2$ and $t_0 = x/(as)$. On the decreasing interval $0 \le t \le \min(1, t_0)$,

$$\beta_s(t) \le 1 - axt \le \mathrm{e}^{-axt}.$$

Consequently,

$$\begin{aligned} |R|_{\mathrm{dec}} &\le \mathrm{e}^{1/2} a^2 m \int_0^\infty t\mathrm{e}^{-raxt}\,\mathrm{d}t = \frac{\mathrm{e}^{1/2} m}{r^2 x^2} \\ &\le \frac{9\mathrm{e}^{1/2}}{mx^2} < \frac{60\delta}{\log A} < \frac{\delta}{1000}. \end{aligned}$$

Here we used $x^2 > \rho/4$, (5.2), and $\log A > L/2 > 2.3 \cdot 10^5$.

If $t_0 < 1$, then β_s is increasing on $[t_0, 1]$, so

$$|R|_{\mathrm{inc}} \le \mathrm{e}^{1/2} a^2 m\, \beta_s(1)^r \le \mathrm{e}^{1/2}(1-\delta) m A^{-c/\delta}, \qquad c := \frac{r}{N} = \frac{m-1}{m+2}.$$

If $\delta \le 1/2$, then $c \ge 9/10$ and

$$|R|_{\mathrm{inc}} \le \mathrm{e}^{1/2} m \left(\frac{m\delta}{2}\right)^{-9/5} \le 4\mathrm{e}^{1/2} L^{9/5} m^{-4/5} < \frac{\delta}{1000}.$$

It remains to consider $\delta \geq 1/2$. Write $\varepsilon = 1 - \delta = a^2$ and $\eta = 3/(m+2)$, so $c = 1 - \eta$. Since $2/\delta \leq 4$ and $c/\delta \leq 2$,

$$\begin{aligned} m\left(\frac{m\delta}{2}\right)^{-c/\delta} &\leq 16m^{1-c/\delta} \\ &= 16m^{(\eta-\varepsilon)/\delta} \\ &\leq 16m^{2\eta}m^{-\varepsilon} \leq 32m^{-\varepsilon}. \end{aligned}$$

The elementary maximum

$$\sup_{\varepsilon \geq 0} \varepsilon m^{-\varepsilon} = \frac{1}{\mathrm{e}\log m}$$

therefore gives

$$|R|_{\mathrm{inc}} \leq \frac{32\mathrm{e}^{1/2}}{\mathrm{e}L} < \frac{20}{L} < \frac{\delta}{1000}.$$

The increasing contribution is absent when $t_0 \geq 1$. Combining the three estimates proves (5.3). ☐

We can now exclude the complementary range.

Proposition 5.3. *There is no counterexample satisfying*

$$m \geq M_\star, \qquad \delta \geq \frac{1}{\log m}, \qquad 0 < a < 1.$$

Proof. By Lemma 5.2,

$$|F(1)| + |R| < \frac{\delta}{100} < 1.$$

If $\mu = 0$, the exact identity (2.5) would give

$$1 \leq |F(1)| + |R| < 1,$$

a contradiction. Hence $\mu \neq 0$. Combining (2.3) and (2.5), we obtain

$$|J| = \frac{m+1}{ma|\mu|}\,|1 - F(1) - R| \geq \frac{m+1}{ma|\mu|}\left(1 - \frac{\delta}{100}\right).$$

Since $|\mu| \leq \sqrt{s} < 1$,

$$|J| > \frac{1 - \delta/100}{a}.$$

But

$$1 - a = \frac{1-a^2}{1+a} = \frac{\delta}{1+a} \geq \frac{\delta}{2} > \frac{\delta}{100},$$

so $1 - \delta/100 > a$. Thus $|J| > 1$, contradicting (2.2). ☐

6. Endpoint cases and completion of the proof

The two remaining values of a are elementary, but we record them because they retain all multiplicities and make the final theorem closed at the endpoints.

Lemma 6.1. *The desired averaged inequality holds when $a = 0$ or $a = 1$. More precisely, if all translated critical points are nonzero, then*

$$\frac{1}{m}\sum_{j=1}^{m} |q_j|^2 \geq 1.$$

If a translated critical point is zero, the conclusion is immediate.

Proof. If $a = 0$, the Gauss–Lucas theorem places all critical points of the original polynomial in $\overline{\mathbb{D}}$; see [RS02, pp. 71–72, Theorem 2.1.1]. Thus the translated critical points satisfy $|w_j| \le 1$, and $|q_j| = |w_j|^{-1} \ge 1$.

Suppose $a = 1$. In the original coordinate write

$$R(u) = (u-1)\prod_{i=1}^{m}(u-u_i), \qquad |u_i| \le 1,$$

and let $c_j = 1 + w_j$ be its critical points. If some $u_i = 1$, then the distinguished zero is multiple and the conclusion is immediate. Otherwise,

$$q_j = \frac{1}{1-c_j},$$

and logarithmic differentiation at $u = 1$ gives

$$\sum_{j=1}^{m} q_j = \frac{R''(1)}{R'(1)} = 2\sum_{i=1}^{m}\frac{1}{1-u_i}.$$

For $|u_i| \le 1$,

$$\operatorname{Re}\frac{1}{1-u_i} \ge \frac{1}{2}.$$

Hence $\operatorname{Re}(m^{-1}\sum_j q_j) \ge 1$, and Cauchy–Schwarz yields

$$\frac{1}{m}\sum_{j=1}^{m}|q_j|^2 \ge \left|\frac{1}{m}\sum_{j=1}^{m} q_j\right|^2 \ge 1.$$

□

We now assemble the ranges.

Proof of Theorem 1.2. Suppose that (1.2) fails. The normalization in Section 2 gives a counterexample with $m = n-1$. Since $n \ge 10^{200000}$,

$$m = n - 1 > 10^{199999} = M_\star.$$

The endpoint cases $a = 0, 1$ are excluded by Lemma 6.1. For $0 < a < 1$, either

$$0 < 1 - a^2 \le \frac{1}{\log m}$$

or

$$1 - a^2 \ge \frac{1}{\log m}.$$

The first alternative is impossible by Proposition 4.1, and the second is impossible by Proposition 5.3. This proves the theorem. □

Proof of Corollary 1.3. Put $u_j = |a - w_j|^{-1}$. By Theorem 1.2, $m^{-1}\sum_j u_j^2 \ge 1$. For $\lambda \ge 2$, the monotonicity of power means gives

$$\left(\frac{1}{m}\sum_{j=1}^{m} u_j^{\lambda}\right)^{1/\lambda} \ge \left(\frac{1}{m}\sum_{j=1}^{m} u_j^{2}\right)^{1/2} \ge 1.$$

□

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